

Part VII: Black Holes and White Holes in VG – From Dynamic Archetype to Cosmic Structure

Abstract: This part establishes the VG perspective on black holes (BH) and white holes (WH) as fundamental dynamic regimes of relational coupling, not as independent field solutions. We introduce the dynamic variable "coupling arrow" and show how the coherent aggregation of these regimes can give rise to both astrophysical phenomena and a possible structure for dark matter.

VII.1 Introduction: Redefining the Problem

In VG, the BH and WH problem is redefined at the level of fundamental relational dynamics. The starting point is not field equations, but the question: what are the possible states of interaction between the primary components (nodes) of reality? BHs and WHs are archetypes of this dynamics.

VII.2 The Dynamic Archetype: The Coupling Arrow and The Two Regimes

We define the central dynamic variable, the coupling arrow $\mathcal{A}_{ij} \in [-1, +1]$, which describes the predominant orientation of vibrational flow between nodes i and j:

1. Basic Definition:

$$\mathcal{A}_{ij} = \eta \langle \vec{S}_i \cdot \vec{S}_j \rangle$$

where η is a positive constant, and $\langle \vec{S}_i \cdot \vec{S}_j \rangle$ is the expected value of the spin correlation operator between nodes. The spin state acts as a "gatekeeper of the intersection."

2. The Two Pure Regimes of Coupling:

- Black Hole (BH) Regime: $\mathcal{A}_{ij} > 0$
 - Interpretation: Predominant vibrational flow from node i to node j . This means a tendency for absorption or relational implosion.
- White Hole (WH) Regime: $\mathcal{A}_{ij} < 0$
 - Interpretation: Predominant vibrational flow from node j to node i . This means a tendency for emission or relational explosion.

3. Nonlocal Coupling and Topological Constraint:

The arrow also influences the nonlocal rigidity term K from the effective action (Part VI). It becomes dependent on the alignment of arrows at neighboring nodes:

$$K_{\text{effective}} = K_0[1 - \delta \mathcal{A}_{ij} \mathcal{A}_{jk}]$$

where δ controls the strength of the constraint. When the arrows are aligned

($\text{product} \mathcal{A}_{ij} \mathcal{A}_{jk} > 0$), $K_{\text{effective}}$ decreases, favoring the isolation and formation of internal domains with coherent orientation. This is the basic mechanism for the topological constraint of a horizon.

VII.3 Black Holes and White Holes in VG: A Debate Resolved

The VG Position: Regimes, Not Isolated Objects

- In VG, BHs and WHs are primarily operational regimes of the fundamental nodal coupling $\hat{J}_{ij}$. They are the "atoms" from which relational dynamics are woven.
- "Mass" as an Emergent Concept: The mass of an astrophysical BH is the emergent binding energy of a huge, coherently oriented sub-network in the BH regime. It is distinct from the energy scale ("Planck mass") of a single fundamental coupling. We "breathe" mini-BH/WH because our ordinary matter is a stable configuration made from them.

VII.4 Consequences: Dark Matter as a Network of WH Relics

The central hypothesis of VG is that stable WH relics (sub-networks locked in a WH regime) are natural candidates for dark matter.

1. Definition of a WH Relic ("Hidden Palace"):

A sub-network with the following configuration:

- Coherent internal coupling: $\bar{J}_{\text{int}}$ large and stable, maintaining internal coherence.
- Extremely weak external coupling: $\kappa_{\text{ext}} \approx 0$, due to the constraint $K_{\text{effective}} \rightarrow 0$ at the interface.
- Locked phase configuration: The phases θ_{ij} of internal couplings are "frozen" in a minimum-energy WH configuration.

2. Emergent Properties of the Relics (VG Dark Matter):

From VG dynamics stem all the essential properties of dark matter:

- Exclusive Gravitational Interaction: The external coupling $\kappa_{\text{ext}} \approx 0$ prevents the exchange of quantum excitations with the environment (photons, particles). However, the sub-network perturbs its background geometry, manifesting only gravitationally.
- Cosmic Stability: The sub-network is locked in a minimum energy plateau of the WH configuration, being practically immutable on a cosmic scale.
- Granular Cloud Behavior: These sub-structures are diffusely distributed in galactic halos, resisting dissipation and explaining galactic rotation profiles.

VII.5 Conclusion and Perspectives

Part VII achieves the conceptual unification between the fundamental dynamic archetype (BH/WH coupling) and cosmic structure. It provides a framework for understanding dark matter not as a cloud of exotic particles, but as hidden architecture – a cosmic population of topological relic structures ("Hidden Palaces") that interact exclusively through geometry.